

3rd Sem- BTech CSE
MID-SEMESTER EXAMINATION, SEPTEMBER, 2024

Course Code: CICPC06 / CBCPC06
Course Title- Design and Analysis of Algorithms

Time- 1.30 hrs.

Max. Marks- 15

Note: - Attempt all the five questions in given sequence. Missing data/ information, if any maybe suitably assumed & mentioned in the answer.

No.	All questions are compulsory.	Marks	CO
Q1			
1a	Define an algorithm and explain the importance of analyzing algorithms.	1	I
1b	What is asymptotic notation? List the common asymptotic notations and its properties.	2	I
Q2			
2a	Given $f(n) = n^2 + n + 1$ and $g(n) = n^2$, prove using theta notation.	2	I
2b	Explain the concept of recursion with an example.	1	I
Q3			
3a	Solve the given sequence with the help of insertion sort. [30,35,15,70,56,23,44,88,79,56,11,13,99].	2	I
3b	Define the divide and conquer paradigm with an example.	1	I
Q4			
4a	Compare AVL trees and Red-Black trees. Which one would be more efficient for search operations and why?	1	II
4b	Provide a detailed analysis of the Heap Sort algorithm, including its time complexity in the best, worst, and average cases.	2	II
Q5			
5a	Construct a binary search tree for the given sequence 45,30,6,8,15,48,60,20,70,12,75,26,36,88,44,14,25,27	2	II
5b	Describe the average-case analysis of the Quick Sort algorithm.	1	II